

Taylor-Informed Indirect Adaptive Predictive Control Using Jacobian-Frozen Affine Predictors

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Abstract—This paper presents a Taylor-informed indirect adaptive predictive control framework for nonlinear sampled-data systems based on Jacobian-frozen affine predictors. The sampled nonlinear dynamics are approximated by a finite Taylor expansion whose polynomial coefficients are identified online using recursive least squares (RLS). At each sampling instant, the Jacobian of the identified map is evaluated at the current operating point and frozen over the prediction horizon to construct an affine predictor for model predictive control. Unlike generic nonlinear feature dictionaries, the implemented polynomial dictionary is a forward-Euler/Taylor-structure-informed reduced dictionary. Exact joint-odd symmetry removes even-total-degree monomials, while additional forward-Euler-informed pruning provides a deliberate model reduction. Numerical simulations on an unstable nonlinear benchmark compare different Taylor degrees and show that higher-order models improve tracking accuracy as the operating point moves farther from the expansion point while maintaining comparable control effort. The complete MATLAB implementation is publicly available to facilitate reproducibility.

I. INTRODUCTION

Indirect adaptive predictive control combining online identification with model predictive control (MPC) enables online adaptation from streaming input–output data while retaining the computational structure of linear predictive control [1]–[3]. At each sampling instant, the predictive model is updated recursively and used to compute the future control sequence over a finite horizon. Recent developments have extended this framework to a growing range of aerospace and robotic applications [4]–[6].

Extending adaptive predictive control to nonlinear systems remains an active research topic. Existing approaches include nonlinear model predictive control, lifting techniques based on observable dictionaries, and adaptive predictive methods employing nonlinear regressors [7]–[10]. More recently, nonlinear extensions of the predictive cost adaptive control (PCAC) framework [11] have been investigated using kernel, polynomial, Fourier, and spline feature dictionaries [12], [13]. One approach is to replace the linear regressor by a richer feature dictionary and continue estimating the corresponding coefficients through recursive least squares (RLS). Polynomial, kernel, and other nonlinear observable representations have therefore been investigated as a means of increasing model expressiveness while preserving linearity in the parameters [14]–[16]. Although these dictionaries often improve approximation capability, they are frequently

selected from general families of basis functions whose connection to the underlying nonlinear plant is indirect.

The present work adopts a different perspective. Rather than introducing a generic nonlinear dictionary, we construct a reduced feature set using the Taylor structure of the sampled-data dynamics together with structural information from a forward-Euler expansion. The resulting monomials have a clear analytical interpretation. Exact symmetries of the sampled flow can justify some eliminations; additional omissions can then be imposed deliberately as forward-Euler-informed model reduction. The latter reduction does not imply that every omitted monomial vanishes in the sampled flow.

When the exact nonlinear basis is known and trusted, it can be used directly. In many applications, however, only partial structural knowledge is available. A Taylor dictionary provides a common local representation without committing to a specific functional basis. As the Taylor degree increases, the same regressor can accommodate richer local nonlinear behavior through coefficient adaptation rather than redesign of the regressor.

After identifying the polynomial coefficients online using RLS, the identified sampled nonlinear map is transformed into a locally affine predictive model by evaluating its Jacobian at the current operating point and freezing the resulting affine coefficients over the prediction horizon. This construction avoids repeated nonlinear propagation while retaining a predictor that adapts online through the identified Taylor coefficients. The resulting predictor is then used within a conventional MPC formulation.

The objective of this paper is to investigate, through numerical simulation, the influence of the Taylor degree on closed-loop tracking performance. In particular, we compare structured Taylor dictionaries of different orders and evaluate the trade-off between model complexity and control performance on a nonlinear benchmark system. To facilitate reproducibility, the complete MATLAB implementation used throughout this study is publicly available at <https://github.com/tamwng/taylor-informed-iapc>.

II. SYSTEM DESCRIPTION

Consider the continuous-time nonlinear system

$$\dot{x} = f(x, u), \quad (1)$$

where $f : \mathbb{R}^n \times \mathbb{R}^m \rightarrow \mathbb{R}^n$, $x \in \mathbb{R}^n$ is the state, and $u \in \mathbb{R}^m$ is the control.

Under zero-order hold with sampling period $T_s > 0$,

$$u(t) = u_k, \quad t \in [kT_s, (k+1)T_s),$$

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where $t \in \mathbb{R}_{\geq 0}$ is continuous time and $k \in \mathbb{N}_0$ is the sampling index. The exact sampled-data dynamics are

$$x_{k+1} = F(x_k, u_k), \quad (2)$$

where $F : \mathbb{R}^n \times \mathbb{R}^m \rightarrow \mathbb{R}^n$ denotes the sampled-data flow map associated with (1).

Let (x_e, u_e) denote a selected equilibrium pair satisfying

$$f(x_e, u_e) = 0. \quad (3)$$

Then,

$$F(x_e, u_e) = x_e.$$

Define the shifted variables

$$\tilde{x} := x - x_e, \quad \tilde{u} := u - u_e.$$

The shifted sampled-data dynamics are

$$\tilde{x}_{k+1} = \bar{F}(\tilde{x}_k, \tilde{u}_k), \quad (4)$$

where

$$\bar{F}(\tilde{x}, \tilde{u}) := F(x_e + \tilde{x}, u_e + \tilde{u}) - x_e. \quad (5)$$

By construction,

$$\bar{F}(0, 0) = 0. \quad (6)$$

The following assumptions are used throughout this paper.

Assumption 1. The state x_k is available for feedback at each time k .

Assumption 2. $\bar{F}$ is well-defined on a neighborhood $\mathcal{N} \subset \mathbb{R}^{n+m}$ of the origin.

Assumption 3. $\bar{F} : \mathcal{N} \rightarrow \mathbb{R}^n$ is of class $\mathcal{C}^{D+1}$ on $\mathcal{N}$.

III. TAYLOR APPROXIMATION OF THE SAMPLED DYNAMICS

Under Assumption 3, $\bar{F} : \mathcal{N} \rightarrow \mathbb{R}^n$ admits a finite Taylor approximation around the origin.

Let

$$p := n + m, \quad z := \begin{bmatrix} \tilde{x} \\ \tilde{u} \end{bmatrix} \in \mathbb{R}^p. \quad (7)$$

Since $\bar{F}(0) = 0$, the order- D Taylor approximation is

$$\bar{F}(z) = Mz + \sum_{d=2}^D F_d(z) + r_D(z), \quad (8)$$

where

$$M := \left. \frac{\partial \bar{F}}{\partial z} \right|_{z=0} \in \mathbb{R}^{n \times p}, \quad (9)$$

and $r_D : \mathcal{N} \rightarrow \mathbb{R}^n$ is the Taylor remainder. For each $d \in \{2, \dots, D\}$, $F_d : \mathbb{R}^p \rightarrow \mathbb{R}^n$ collects all homogeneous polynomial terms of total degree d . Thus, F_2 contains all quadratic terms, F_3 contains all cubic terms, and so forth. For $D = 1$, the summation in (8) is empty.

Equation (8) shows that the local Taylor approximation is a polynomial in the components of z . Define the full Taylor index set

$$\mathcal{A}_D := \{\alpha \in \mathbb{N}_0^p : 1 \leq |\alpha| \leq D\}, \quad |\alpha| := \sum_{j=1}^p \alpha_j. \quad (10)$$

For each $\alpha \in \mathcal{A}_D$, define the monomial

$$z^\alpha := \prod_{j=1}^p z_j^{\alpha_j}, \quad (11)$$

where z_j is the j -th component of z .

Let $\mathcal{A}_D = \{\alpha_1, \dots, \alpha_q\}$. The polynomial regressor is

$$\phi_D(z) := [z^{\alpha_1} \quad \dots \quad z^{\alpha_q}]^\top \in \mathbb{R}^q. \quad (12)$$

The regressor ϕ_D contains all pure and crossed monomials of total degree from 1 to D . Therefore,

$$q = \binom{p+D}{D} - 1. \quad (13)$$

Accordingly, the order- D Taylor model is written compactly as

$$\bar{F}(z) = \Theta \phi_D(z) + r_D(z), \quad (14)$$

where $\Theta \in \mathbb{R}^{n \times q}$ contains the Taylor coefficients.

IV. RECURSIVE TAYLOR COEFFICIENT IDENTIFICATION

The Taylor model (14) is linear in the coefficient matrix Θ . Neglecting the Taylor remainder, identifying the sampled dynamics reduces to estimating the coefficient matrix Θ .

For each $k \geq 0$, define

$$z_k := \begin{bmatrix} \tilde{x}_k \\ \tilde{u}_k \end{bmatrix}, \quad \phi_k := \phi_D(z_k). \quad (15)$$

The a priori one-step prediction of $\tilde{x}_{k+1}$ is

$$\hat{\tilde{x}}_{k+1} = \hat{\Theta}_k \phi_k. \quad (16)$$

Vectorize the coefficient matrix as

$$\theta_k := \text{vec}(\hat{\Theta}_k) \in \mathbb{R}^{nq}, \quad (17)$$

and define the regression matrix

$$\varphi_k := \phi_k^\top \otimes I_n \in \mathbb{R}^{n \times nq}. \quad (18)$$

Then (16) becomes

$$\hat{\tilde{x}}_{k+1} = \varphi_k \theta_k. \quad (19)$$

The prediction error is

$$e_k := \tilde{x}_{k+1} - \varphi_k \theta_k. \quad (20)$$

Using standard RLS with forgetting factor $\lambda \in (0, 1]$ and $P_0 \succ 0$, define

$$L_k = \lambda^{-1} P_k, \quad (21)$$

$$P_{k+1} = L_k - L_k \varphi_k^\top (I_n + \varphi_k L_k \varphi_k^\top)^{-1} \varphi_k L_k, \quad (22)$$

$$\theta_{k+1} = \theta_k + P_{k+1} \varphi_k^\top e_k. \quad (23)$$

The pair (P_{k+1}, θ_{k+1}) is the posterior RLS estimate obtained after processing the measured transition $(z_k, \tilde{x}_{k+1})$. In the numerical study, these one-step transitions are generated by the zero-order-hold RK4 simulation of the continuous-time plant; forward Euler is not used to propagate the simulated plant. Thus, once $\tilde{x}_{k+1}$ is measured, θ_{k+1} is available for the subsequent control synthesis.

Using this posterior estimate, reshape θ_{k+1} into $\hat{\Theta}_{k+1} \in \mathbb{R}^{n \times q}$ and define the identified sampled Taylor map

$$\hat{F}_{k+1}(z) := \hat{\Theta}_{k+1} \phi_D(z). \quad (24)$$

Thus, $\hat{F}_{k+1}$ is obtained from (14) by replacing the unknown coefficient matrix Θ with its RLS estimate $\hat{\Theta}_{k+1}$ and neglecting the Taylor remainder.

The map $\hat{F}_{k+1}$ is nonlinear in z but linear in the identified coefficients. This structure enables direct RLS identification. Other RLS variants, including variable-rate forgetting or covariance reset, can be used without changing the Jacobian-frozen construction below.

In the sequel, $\hat{F}_k$ denotes the most recent identified map available at the current control instant.

V. JACOBIAN-FROZEN AFFINE PREDICTOR

The identified Taylor map $\hat{F}_k$ is nonlinear in the shifted state and input. Exact finite-horizon propagation of $\hat{F}_k$ would lead to nonlinear predictive control. Instead, we construct a local affine predictor by freezing the Jacobian of $\hat{F}_k$ at the current operating point.

A. Operating Point

At step k , the shifted state $\tilde{x}_k$ is measured and $\tilde{u}_k$ is the currently applied input. The control computation produces the subsequent input $\tilde{u}_{k+1}$. Accordingly, the operating point is chosen as

$$\bar{z}_k := z_k = \begin{bmatrix} \tilde{x}_k \\ \tilde{u}_k \end{bmatrix} \in \mathbb{R}^{n+m}. \quad (25)$$

The initial input u_0 is prescribed.

The Taylor approximation remains centered at the equilibrium $z = 0$. The operating point $\bar{z}_k$ is used only for Jacobian evaluation.

B. Jacobian Compilation

Compute the Jacobian of the identified Taylor map at $\bar{z}_k$:

$$J_k := \left. \frac{\partial \hat{F}_k}{\partial z} \right|_{z=\bar{z}_k} \in \mathbb{R}^{n \times (n+m)}. \quad (26)$$

Partition

$$J_k = \begin{bmatrix} A_k & B_k \end{bmatrix}, \quad (27)$$

where $A_k \in \mathbb{R}^{n \times n}$ and $B_k \in \mathbb{R}^{n \times m}$ are the Jacobians of $\hat{F}_k$ with respect to the shifted state and input, respectively.

The first-order local approximation of $\hat{F}_k$ at $\bar{z}_k$ is

$$\hat{F}_k(z) \approx \hat{F}_k(\bar{z}_k) + J_k(z - \bar{z}_k). \quad (28)$$

Substituting $z = [\tilde{x}^\top \tilde{u}^\top]^\top$ and partitioning J_k according to (27) yields

$$\hat{F}_k(z) \approx c_k + A_k \tilde{x} + B_k \tilde{u}, \quad (29)$$

where

$$c_k := \hat{F}_k(\bar{z}_k) - A_k \tilde{x}_k - B_k \tilde{u}_k. \quad (30)$$

The Jacobian-frozen predictive predictor is therefore

$$\tilde{x}_{i+1|k} = c_k + A_k \tilde{x}_{i|k} + B_k \tilde{u}_{i|k}, \quad i = 0, \dots, N-1. \quad (31)$$

The matrices A_k , B_k , and c_k are frozen over the horizon.

C. Reduction to Linear PCAC

The proposed scheme contains linear PCAC as its first-order special case.

Corollary 1: If $D = 1$, then the Jacobian-frozen predictor (31) reduces to the linear PCAC propagation model.

Proof: For $D = 1$, the Taylor regressor satisfies $\phi_1(z) = z$. Hence,

$$\hat{F}_k(z) = \hat{\Theta}_k z, \quad (32)$$

where $\hat{\Theta}_k \in \mathbb{R}^{n \times (n+m)}$. Therefore,

$$J_k = \hat{\Theta}_k = \begin{bmatrix} A_k & B_k \end{bmatrix} \quad (33)$$

is independent of the operating point. Moreover,

$$c_k = \hat{F}_k(\bar{z}_k) - J_k \bar{z}_k \quad (34)$$

$$= \hat{\Theta}_k \bar{z}_k - \hat{\Theta}_k \bar{z}_k = 0. \quad (35)$$

Substituting $c_k = 0$ into (31) yields

$$\tilde{x}_{i+1|k} = A_k \tilde{x}_{i|k} + B_k \tilde{u}_{i|k}, \quad (36)$$

which is the linear PCAC propagation model. ■

D. Elimination of Factor-Freezing Ambiguity

The Jacobian freeze avoids arbitrary choices in mixed nonlinear terms. For example, consider

$$F(y, u) = \alpha y + \beta u + au^2 + bu y + cy^2. \quad (37)$$

At $(\bar{y}, \bar{u})$, the Jacobian-frozen predictor is

$$F(y, u) \approx C_{\text{loc}} + A_{\text{loc}} y + B_{\text{loc}} u, \quad (38)$$

where

$$A_{\text{loc}} = \alpha + b\bar{u} + 2c\bar{y}, \quad (39)$$

$$B_{\text{loc}} = \beta + 2a\bar{u} + b\bar{y}, \quad (40)$$

$$C_{\text{loc}} = F(\bar{y}, \bar{u}) - A_{\text{loc}} \bar{y} - B_{\text{loc}} \bar{u}. \quad (41)$$

Thus, the cross term satisfies

$$uy \approx \bar{u} y + \bar{y} u - \bar{u} \bar{y}. \quad (42)$$

This approximation matches the value and Jacobian at the operating point. It does not require choosing whether to freeze u or y .

VI. PREDICTIVE CONTROL SYNTHESIS

Following the PCAC formulation [11], the Jacobian-frozen predictor (31) is used over a prediction horizon N .

At step k , initialize

$$\tilde{x}_{0|k} = \tilde{x}_k, \quad \tilde{u}_{0|k} = \tilde{u}_k. \quad (43)$$

The predicted states satisfy

$$\tilde{x}_{i+1|k} = c_k + A_k \tilde{x}_{i|k} + B_k \tilde{u}_{i|k}, \quad i = 0, \dots, N-1. \quad (44)$$

Here, $\tilde{u}_{0|k}$ is known, while $\tilde{u}_{1|k}, \dots, \tilde{u}_{N-1|k}$ are decision variables. This indexing follows the PCAC convention, where $\tilde{u}_{1|k}$ is applied at step $k+1$.

Let

$$\Delta \tilde{u}_{i|k} := \tilde{u}_{i|k} - \tilde{u}_{i-1|k}, \quad i = 1, \dots, N-1, \quad (45)$$

and let $\tilde{r}_{i|k}$ denote the reference in shifted coordinates.

Assume that the state and input constraint sets are the convex polyhedra

$$\mathcal{X} = \{\tilde{x} \in \mathbb{R}^n : H_x \tilde{x} \leq h_x\}, \quad (46)$$

$$\mathcal{U} = \{\tilde{u} \in \mathbb{R}^m : H_u \tilde{u} \leq h_u\}. \quad (47)$$

To soften the state constraints, introduce $\varepsilon_{i|k} \geq 0$ and solve

$$\begin{aligned} & \underset{\substack{\tilde{x}_{1|k}, \dots, \tilde{x}_{N|k}, \\ \tilde{u}_{1|k}, \dots, \tilde{u}_{N-1|k}, \\ \varepsilon_{1|k}, \dots, \varepsilon_{N|k}}}{\text{minimize}} \quad \frac{1}{2} \sum_{i=1}^N \|\tilde{x}_{i|k} - \tilde{r}_{i|k}\|_{Q_i}^2 \\ & + \frac{1}{2} \sum_{i=1}^{N-1} \|\Delta \tilde{u}_{i|k}\|_R^2 + \frac{1}{2} \sum_{i=1}^N \|\varepsilon_{i|k}\|_S^2 \quad (48) \end{aligned}$$

subject to (44),

$$H_x \tilde{x}_{i|k} \leq h_x + \varepsilon_{i|k}, \quad i = 1, \dots, N,$$

$$H_u \tilde{u}_{i|k} \leq h_u, \quad i = 1, \dots, N-1,$$

$$\varepsilon_{i|k} \geq 0, \quad i = 1, \dots, N,$$

where $Q_i \succeq 0$, $R \succ 0$, $S \succ 0$, and $\|v\|_Q^2 := v^\top Q v$.

Since A_k , B_k , and c_k are fixed over the horizon, (48) is a convex quadratic program. The slack variables prevent infeasibility caused solely by the state constraints, but they also allow the nominal state constraint to be violated. No recursive-feasibility or closed-loop stability guarantee is asserted because the identified model and its Jacobian-frozen predictor vary online.

Let $\tilde{u}_{1|k}^*$ denote the first optimized input. The computed control is

$$u_{k+1} = u_e + \tilde{u}_{1|k}^*. \quad (49)$$

Thus, the measured transition from k to $k+1$ is generated using u_k ; the newly computed u_{k+1} is held over the subsequent sampling interval.

VII. CASE STUDY: UNSTABLE TRIGONOMETRIC PLANT

A. Plant and Sampled Dynamics

Consider the continuous-time scalar system

$$\dot{x} = f(x, u) := a \sin x \cos x + b \cos^2 x \sin u, \quad (50)$$

where $a > 0$ and $b > 0$. Select $(x_e, u_e) = (0, 0)$ as the equilibrium. Hence, $\tilde{x} = x$ and $\tilde{u} = u$, and tildes are omitted throughout this section. Near the origin,

$$\dot{x} = ax + bu + \mathcal{O}(\|(x, u)\|^3), \quad (51)$$

so the equilibrium is open-loop unstable and its linearization is controllable.

Under zero-order hold with sampling period $T_s = 0.05$ s, let F denote the exact sampled-data flow map. In the numerical study, the continuous-time plant is propagated over every sampling interval by an RK4 step with the input held constant. The RLS estimator identifies the resulting one-step transitions. Forward Euler is not used to propagate the simulated plant.

For analytical reference only, a forward-Euler step applied to (50) gives

$$\begin{aligned} x_{k+1} &= F_{FE}(x_k, u_k) \\ &:= x_k + T_s (a \sin x_k \cos x_k + b \cos^2 x_k \sin u_k). \end{aligned} \quad (52)$$

and its linear expansion near the origin is

$$x_{k+1} = (1 + aT_s)x_k + bT_s u_k + \mathcal{O}(\|(x_k, u_k)\|^3), \quad (53)$$

whose open-loop eigenvalue is $1 + aT_s > 1$.

Since

$$f(-x, -u) = -f(x, u), \quad (54)$$

uniqueness of solutions implies that the exact sampled flow satisfies the joint-odd symmetry

$$F(-x, -u) = -F(x, u).$$

The same identity holds exactly for the implemented RK4 one-step map because each RK4 stage preserves this joint sign change. Consequently, the sampled flow and the implemented RK4 map have no even-total-degree terms in their Taylor expansions about the origin.

B. Taylor Structure and Coefficient Pruning

Let

$$z = [x \quad u]^\top. \quad (55)$$

As an analytical reference, expanding the forward-Euler expression (52) about the origin through degree five yields

$$\begin{aligned} F_{FE}(x, u) &= (1 + aT_s)x + bT_s u \\ &\quad - \frac{2aT_s}{3}x^3 - bT_s x^2 u - \frac{bT_s}{6}u^3 \\ &\quad + \frac{2aT_s}{15}x^5 + \frac{bT_s}{3}x^4 u + \frac{bT_s}{6}x^2 u^3 + \frac{bT_s}{120}u^5 \\ &\quad + \mathcal{O}(\|z\|^7). \end{aligned} \quad (56)$$

TABLE I
NUMBER OF ESTIMATED COEFFICIENTS IN THE FULL AND
IMPLEMENTED REDUCED DICTIONARIES.

Degree D	Full dictionary	Reduced dictionary
1	2	2
3	9	5
5	20	9
7	35	14

Joint-odd symmetry removes all even-total-degree terms exactly. The implemented dictionary then applies an additional, deliberate model reduction informed by the forward-Euler structure: for a degree- D truncation, it retains only monomials of the form

$$x^{2i+1}, \quad x^{2i}u^{2j+1}, \quad i, j \in \mathbb{N}_0, \quad (57)$$

with total degree at most D . This additional pruning does not assert that every omitted odd-total-degree monomial vanishes in the RK4 sampled flow. The resulting feature set is therefore termed a forward-Euler/Taylor-structure-informed reduced dictionary.

The resulting regressors are

$$\begin{aligned} \varphi_1^{\text{FE}} &= [x \quad u]^\top, \\ \varphi_3^{\text{FE}} &= [x \quad u \quad x^3 \quad x^2u \quad u^3]^\top, \\ \varphi_5^{\text{FE}} &= [x \quad u \quad x^3 \quad x^2u \quad u^3 \quad x^5 \quad x^4u \quad x^2u^3 \quad u^5]^\top, \\ \varphi_7^{\text{FE}} &= [(\varphi_5^{\text{FE}})^\top \quad x^7 \quad x^6u \quad x^4u^3 \quad x^2u^5 \quad u^7]^\top. \end{aligned}$$

Table I compares the number of coefficients in the full Taylor dictionary and the implemented reduced dictionary.

For analytical reference, (56) gives the coefficient vector associated with φ_5^{FE} as

$$\theta_5^* = \begin{bmatrix} 1+aT_s \\ bT_s \\ -2aT_s/3 \\ -bT_s/6 \\ 2aT_s/15 \\ bT_s/3 \\ bT_s/6 \\ bT_s/120 \end{bmatrix}. \quad (58)$$

Thus, the forward-Euler-informed model reduction reduces the number of estimated coefficients from 20 to 9 for $D = 5$, and from 35 to 14 for $D = 7$. The vector θ_5^* is an analytical forward-Euler reference only; it is neither used by the controller nor numerically compared with the identified RK4-transition coefficients. The controller estimates the retained coefficients online using RLS.

No constant monomial is estimated; the affine term arises subsequently from Jacobian freezing.

C. Identified Map and Jacobian-Frozen Predictor

For each Taylor degree $D \in \{1, 3, 5, 7\}$, let φ_D^{FE} denote the implemented reduced regressor defined by the monomial pattern above. The identified sampled map is

$$\hat{F}_{D,k}(x, u) = \hat{\theta}_{D,k}^\top \varphi_D^{\text{FE}}(x, u), \quad (59)$$

where $\hat{\theta}_{D,k}$ is updated online using RLS.

At each step k , the Jacobian-frozen coefficients are evaluated at the current operating point (x_k, u_k) as

$$A_k = \left. \frac{\partial \hat{F}_{D,k}}{\partial x} \right|_{(x_k, u_k)}, \quad (60)$$

$$B_k = \left. \frac{\partial \hat{F}_{D,k}}{\partial u} \right|_{(x_k, u_k)}, \quad (61)$$

$$c_k = \hat{F}_{D,k}(x_k, u_k) - A_k x_k - B_k u_k. \quad (62)$$

For the fifth-order reduced regressor φ_5^{FE} , let $\hat{\theta}_{ij,k}$ denote the estimated coefficient multiplying $x^i u^j$. The resulting Jacobian coefficients are

$$\begin{aligned} A_k &= \hat{\theta}_{10,k} + 3\hat{\theta}_{30,k}x_k^2 + 2\hat{\theta}_{21,k}x_k u_k \\ &\quad + 5\hat{\theta}_{50,k}x_k^4 + 4\hat{\theta}_{41,k}x_k^3 u_k + 2\hat{\theta}_{23,k}x_k u_k^3, \end{aligned} \quad (63)$$

$$\begin{aligned} B_k &= \hat{\theta}_{01,k} + \hat{\theta}_{21,k}x_k^2 + 3\hat{\theta}_{03,k}u_k^2 \\ &\quad + \hat{\theta}_{41,k}x_k^4 + 3\hat{\theta}_{23,k}x_k^2 u_k^2 + 5\hat{\theta}_{05,k}u_k^4. \end{aligned} \quad (64)$$

The Jacobian-frozen predictor is therefore

$$x_{i+1|k} = c_k + A_k x_{i|k} + B_k u_{i|k}, \quad i = 0, \dots, N-1. \quad (65)$$

For analytical reference only, the Jacobian of the forward-Euler expression (52) is

$$A_{\text{FE}}(x, u) = 1 + T_s [a \cos(2x) - b \sin(2x) \sin u], \quad (66)$$

$$B_{\text{FE}}(x, u) = T_s b \cos^2 x \cos u. \quad (67)$$

VIII. NUMERICAL EVALUATION

This section evaluates the effect of the Taylor degree on closed-loop tracking performance. The continuous-time plant (50) is simulated under zero-order hold using fourth-order Runge-Kutta (RK4) with $T_s = 0.05$ s over each sampling interval. RLS identifies the one-step transitions generated by this RK4 simulation. The controller uses the forward-Euler/Taylor-structure-informed reduced dictionary and identified map and the Jacobian-frozen predictor developed above. All cases $D \in \{1, 3, 5, 7\}$ use the same initialization data, constraints, prediction horizon, and control weights.

The complete MATLAB implementation used throughout this study is publicly available at <https://github.com/tamwng/taylor-informed-iapc>.

A. Simulation Setup

The numerical configuration is summarized in Table II. During the first N_{id} samples, a fixed multilevel input sequence is applied and the RLS estimator is updated. The multiple initialization levels excite the odd monomials retained in the reduced dictionary. The predictive controller is activated at $k = N_{\text{id}}$. The state bound $|x_k| \leq 0.75$ is softened using the slack variables in (48); it may therefore be violated. The input bound is hard.

After the initialization phase, the reference is an amplitude-swept sinusoid. For the j th segment, the reference is defined by

$$r_k = A_j \sin(2\pi f(k - N_{\text{id}})T_s),$$

TABLE II
NUMERICAL CONFIGURATION.

Quantity	Value
Plant parameters	$a = 0.5, b = 1.5$
Sampling period	$T_s = 0.05$ s
Plant integration	RK4 with ZOH input
Initial state	$x_0 = 0$
Constraints	soft $ x_k \leq 0.75$, hard $ u_k \leq 1$
Taylor degrees	$D \in \{1, 3, 5, 7\}$
Prediction horizon	$N = 8$
Forgetting factor	$\lambda = 1$
Initial RLS estimate	$\hat{\theta}_{D,0} = 0_{q_D^{FE}}$
Initial covariance	$P_0 = 10^5 I_{q_D^{FE}}$
State weights	$Q_i = 1, i < N, Q_N = 10$
Input-increment weight	$R = 5 \times 10^{-2}$
State-slack weight	$S = 10^5$
Initialization length	$N_{id} = 60$ samples
Initialization input	$u_k \in \{0, \pm 0.15, \pm 0.30, \pm 0.45, \pm 0.60\}$
Reference type	Amplitude-swept sinusoid
Segment length	$K_r = 300$ samples
Sinusoidal frequency	0.10 Hz
Amplitudes	$\{0.25, 0.55, 0.80, 0.90, 0\}$

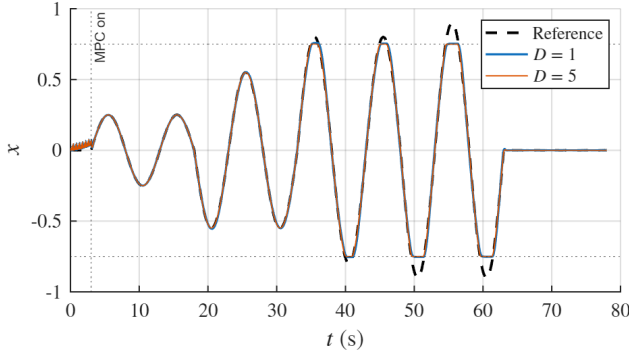


Fig. 1. Closed-loop state tracking for the first- and fifth-order Taylor models. The horizontal dotted lines indicate the nominal bound $|x| \leq 0.75$, which is softened with slack variables and may be violated.

where $f = 0.10$ Hz and

$$A_j \in \{0.25, 0.55, 0.80, 0.90, 0\}.$$

Each amplitude is maintained for $K_r = 300$ samples before transitioning to the next level. The increasing amplitudes progressively excite higher-order nonlinearities. Because the state constraint is soft, the closed-loop state can exceed its nominal bound.

The cases are compared using state trajectories, input trajectories, tracking errors, and quantitative performance metrics computed over each amplitude segment.

B. Closed-Loop Results

Figures 1–3 summarize the closed-loop responses obtained with the proposed controller. Figures 1 and 2 show the state trajectories and logarithmic tracking errors, respectively. Each per-segment metric uses its 300-sample half-open interval $[\text{segmentStart}, \text{segmentStop})$; the evaluated post-transient MAE in Figure 3 discards the first 200 samples and averages the remaining 100. The overall RMSE and MAE in Table III use stored samples $k \geq N_{id}$ through and including

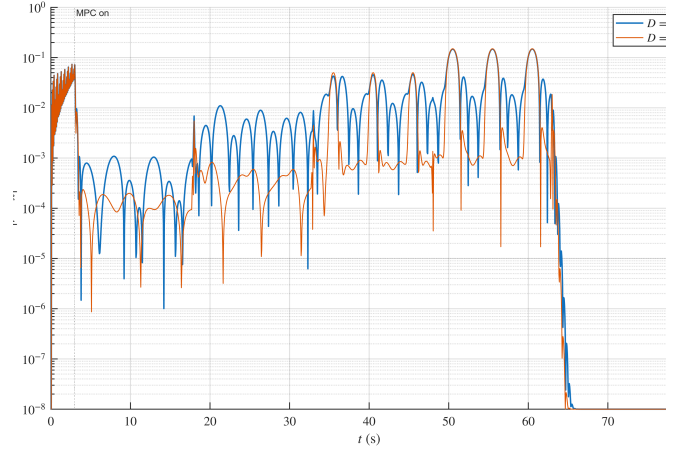


Fig. 2. Absolute tracking error on a logarithmic vertical scale for the first- and fifth-order Taylor models. The plot includes the segment transients.

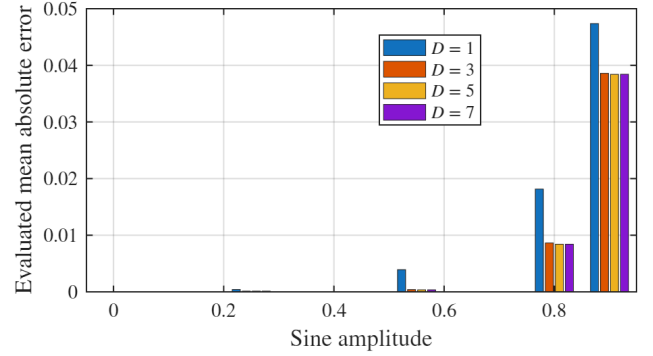


Fig. 3. Evaluated post-transient mean absolute error (MAE) for each reference amplitude and Taylor degree. For every 300-sample amplitude segment, the first sinusoidal period (200 samples) is discarded and the MAE is evaluated over the remaining 100 samples.

the final sample $k = K$. The root-mean-square error (RMSE) measures the overall tracking accuracy, the mean absolute error (MAE) measures the average tracking error, the maximum state violation reports the largest exceedance of the nominal soft state bound, and the total input variation (I.V.) quantifies the accumulated variation of the control input.

Figure 1 compares the responses obtained with the first- and fifth-order Taylor models. Both responses track the reference, with visible clipping near the softened state bound during the larger-amplitude segments. The nonzero violations in Table III confirm that exact state-constraint satisfaction is not claimed.

Figure 2 presents the corresponding absolute tracking errors on a logarithmic vertical scale, including the segment transients. The logarithmic scale displays both the small within-segment errors and the larger error peaks in the same panel.

The trend is summarized quantitatively in Figure 3, which reports the evaluated post-transient MAE for each sinusoidal amplitude after the 200-sample discard. This metric is distinct from the overall MAE in Table III. At small amplitudes, the differences between Taylor degrees are negligible. As

TABLE III
CLOSED-LOOP OVERALL PERFORMANCE FOR DIFFERENT TAYLOR
DEGREES D .

D	RMSE	MAE	Maximum state violation	Total input variation
1	0.03199	0.01418	7.37×10^{-3}	12.41
3	0.03092	0.00971	3.50×10^{-4}	11.75
5	0.03090	0.00958	2.56×10^{-4}	11.86
7	0.03089	0.00959	3.52×10^{-4}	11.81

the amplitude increases, the first-order model exhibits consistently larger mean errors than the higher-order models, whereas the third, fifth, and seventh-order models remain close to one another. This behavior is consistent with the fact that higher-order nonlinear terms become increasingly influential as the system operates farther from the equilibrium about which the Taylor expansion is constructed.

IX. CONCLUSION

This paper presented a Taylor-informed indirect adaptive predictive control framework based on Jacobian-frozen affine predictors for nonlinear systems. In the numerical study, the continuous-time plant is simulated under zero-order hold using RK4 with $T_s = 0.05$ s, and RLS identifies the resulting one-step transitions. The implemented forward-Euler/Taylor-structure-informed reduced dictionary is identified online using recursive least squares (RLS), yielding an adaptive representation of the sampled nonlinear map. At each sampling instant, the Jacobian of the identified map is evaluated at the current operating point and frozen over the prediction horizon, producing an affine predictor suitable for finite-horizon model predictive control.

The reported results show lower overall MAE for $D \in \{3, 5, 7\}$ than for $D = 1$, with similar total input variation. Joint-odd symmetry removes even-total-degree monomials exactly. The additional monomial pruning is a deliberate forward-Euler-informed model reduction; no claim is made that every omitted monomial vanishes in the RK4 sampled flow. The softened state constraint can be violated, as reflected by the reported maximum violations.

Future work will investigate variable-rate forgetting for improved adaptation under changing operating conditions, extend the proposed framework to systems with time delays [17], and validate the approach on real-world physical platforms.

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